

Visual Augmentation of Audio Semantics for Accessibility

Technical report

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Abstract

[Write last. One paragraph: the task (show a deaf viewer the sounds whose source is not on screen), the pipeline (seven stages of off-the-shelf models, no training), the benchmark (274 clips, four categories), the evaluation (VLM-LLM protocol with an independent reference), the headline (`v3_indep` table), and the two findings that survive every reference: the gate is bounded by what a VLM can answer about visibility, and the automatic reference must be built by models that never saw the system’s output.]

1 Introduction

[From notes §The Accessibility Gap.]

1.1 Problem statement

Given a video, produce the same video with a side panel that shows, for the duration of each ambient sound whose source is not visible on screen, a picture of that sound’s event – and shows nothing otherwise. Speech and music are excluded. No model is trained; the system is an inference pipeline over open-source foundation models, and human involvement is limited to the benchmark labels used for evaluation.

1.2 Research questions

1. Which types of audio information contribute most to scene understanding when presented visually?
2. What level of semantic granularity should be used to represent ambient sounds?
3. Can automatically generated visual augmentations improve accessibility beyond subtitles?
4. How should semantic audio augmentation systems be evaluated?
5. How does the proposed pipeline compare with representative audio-to-visual approaches?

1.3 Contributions

- Formulation of semantic audio augmentation as an accessibility task, with a precise definition of when a picture is due (sound heard, source not visible, not speech or music).
- A seven-stage inference pipeline of off-the-shelf models (§3).
- A benchmark of 274 human-labelled clips in four scenarios (§4).

- An automatic evaluation protocol, and the finding that its reference must be independent of the system under test (§5).
- An experimental comparison with two baselines, with paired statistics and an oracle bound (§6).

2 Related work

[Condense docs/literature_review.tex; one paragraph per area, positioned against this project.]

3 Method

3.1 Pipeline

stage	model	role
1 audio extraction	FFmpeg	16 kHz mono wav
2 video understanding	OWLv2 (cheap pass)	candidate visible source
3 speech recognition	Whisper (configurable)	transcript, used only a
4 audio event detection	BEATs iter3+ AS2M (527 AudioSet classes)	events with onsets
5 cross-modal reasoning	Qwen2.5-VL-7B	place, visibility, kind, c
6 visual augmentation	FLUX.1-schnell, 768 px	one picture per sound,
7 evaluation	Qwen2.5-VL / Mistral-7B judge + independent reference	§5

3.2 Stage 4: detection and onset timing

Detector. Stage 4 tags audio with BEATs (iter3+, fine-tuned on AudioSet-2M; 0.486 mAP against PANNs CNN14’s 0.431 on the same 527 labels). BEATs is a clip-level tagger, so a 2 s window is scored every 0.25 s and shaped like frame-wise output; a reflected lead-in pad lets a sound in the first two seconds be heard, and a window’s score is stamped 0.5 s before its end so that an onset is never early. The swap from PANNs was made after a review of eight demo clips traced eleven of fourteen complaints to the detector’s labels (a siren as “truck horn”, a crying baby as “sheep”); on the same clips every one resolved. Speech and music classes are never shown. The display bar is 0.35, set once on the development split and not moved since (§3.6).

Onset. A window cannot say where inside it a sound begins, and under the plain stamp every onset on three hand-checked clips was late by +0.25 to +2.0 s. Two standard pieces on the same model fix most of it. *Hysteresis*: a sound must reach the bar, then extends back through any contiguous stretch above half of it (an approaching helicopter scores 0.15–0.42 for 1.5 s before crossing 0.35). *Prefix-silencing occlusion* inside the first window that fired: silence its opening t seconds for 26 values of t , score them in one batched forward pass, and read the onset as the first cut that removes a tenth of the class’s evidence in logit space. It is class-conditional, so a voice in the same window does not pull it. Three alternatives were measured and rejected: finer windows and frame-level detectors (PANNs 10 ms, BEATs-strong 40 ms) miss about half the sounds; a class-activation map over BEATs’ 160 ms tokens moves every onset by exactly 0 or −1.5 s, because the model was fine-tuned only through the token mean and no token was ever asked to be local. Results in §6.

3.3 Stage 5: is the source on screen?

The proposal’s premise is that a picture is due only when the source of a sound is *not* on screen, so this question carries the system. It was first answered by a hand-written dictionary of about

thirty labels mapped to CLIP prompts; a sound outside the dictionary could never be marked visible, so laughter, footsteps and a telephone were structurally ungated. It is now asked at run time about the sound that was actually detected, from frames sampled across that sound’s own span.

Name it, do not confirm it. The VLM (Qwen2.5-VL-7B) is asked to name the thing in the frames making that sound, or “nothing”. A yes/no question to a VLM collects agreement rather than evidence; a name can be checked. What is in the frame is perceptual and is asked of the pixels; whether that thing makes this sound is world knowledge and is asked as plain text, so a loosely related object cannot count as the source without a table of which objects make which sounds.

The action, not the object. A baby in its mother’s arms was named as the source of the crying and silenced; a deaf viewer looking at it learns nothing about the sound. A second question therefore asks whether the thing can be seen *making the sound now*. Three signals vote (open naming, the world-knowledge check, an open description of the frames); every two-way question is asked in both option orderings with agreement required, because the 7B model prefers a position (it answered “(b)” in 22 of 23 calls whichever option (b) was). Visibility is judged per 5s stretch, but one continuous sound gets one continuous picture even if its source enters the frame midway.

Speech as gate context. For each sound with speech within three seconds, one text-only forced choice asks whether the people on the soundtrack are reacting to it. A yes puts the sound first in line when the panel is full and can rescue a sound just under the bar; it never reaches a picture and never overrides visibility. It is not asked for fewer than four transcribed words, since Whisper transcribes non-speech as phonetic junk. On the 100-clip run it was asked 77 times and answered yes twice.

The frames may choose between the detector’s own candidates. When two labels share both boundaries within 0.6s and are not ancestor and descendant (one sound with two names, from the same windows), the VLM is shown the frames and asked which it is, with “cannot tell” keeping both. The audio proposes, the video picks, and nothing is added that the audio did not hear.

3.4 Stage 5: one picture per source

The detector emits label families (*Laughter*, *Snicker*, *Chuckle*) and a clip once produced three near-identical pictures of people laughing. Three general mechanisms were tried. Label embeddings fail because single words embed too tightly (*Dog/Cat* 0.90 above *Laughter/Snicker* 0.80). Asking the VLM whether two depictions show one thing fails because it answers the question literally: “different” for 90% of pairs, including an owl on a branch against an owl’s hoot. A similarity threshold fails because it must be re-tuned whenever a prompt elsewhere changes, which happened three times. What works is the detector’s own taxonomy: the AudioSet ontology (CC BY 4.0), vendored as child→parent edges, and the rule that one label must be an *ancestor* of the other – a shared parent is not enough, since *Animal* would swallow *Dog* with *Sheep*. The rule is 14/14 on every pair the demos produced, with no threshold. A family label explains a member only where they overlap in time (interval subtraction), so an all-clip *Bird* does not swallow a five-second *Owl*. Depiction similarity survives as a secondary signal at 0.80, and only for pairs that overlap in time.

3.5 Stage 6: what the picture shows

Depiction. For each sound the VLM writes a two-to-five word event phrase (“glass shattering”, “audience clapping”), built from the detector’s label, the sound’s kind as seen in the frames and a *place* of at most four words asked for separately. A place replaced the one-sentence scene description after that sentence’s objects were glued onto every sound: an *Air horn* became “air

horn blaring from silver van” and rendered as a picture of a van. The transcript never reaches the prompt: when it did, a *Hiccup* became “Hiccup on the phone, slow down, how many”. No example phrases are given; the prompt is built from the detector, the audio and the video only.

Validation by choice. A yes/no check approved “man on motorcycle drives past silver van” as a picture of *Laughter*. A comparative check (is the phrase closer to the label than the scene sentence was?) was right on eight cases and 11/17 on seventeen, because it measured phrase length. The working check asks the VLM to *choose*: given the phrase and the other sounds detected in the same clip plus “none of them”, which sound does it show, asked in both orderings.

Rendering and panel. FLUX.1-schnell renders the phrase at 768px on a plain white background (chosen over PixArt- Σ on twelve real prompts). The panel is divided by simultaneity, not inventory: rows are the largest number of sounds heard at once (at most three), a picture is held at least 1.5s, bursts closer than 0.8s are one appearance, and a cell is empty when its sound is absent. The first design parked every sound’s picture beside the video for the whole clip, dimmed when silent; a panel that always looks the same carries no information, and a picture on screen while its sound is absent makes a claim a deaf viewer cannot check.

3.6 Design process

Every earlier fix had been made on the clip in front of us, and each knob had been recalibrated by the next clip. Since 2026-09-13 the rules are: thresholds are set once on the development split; every failure seen in a demo is logged in a failure catalogue with clip, stage and confidence; a design change needs three entries of one pattern or a metric move on the development split; demos illustrate and never tune; and every change is general and re-measured. The catalogue holds 18 entries, seven fixed as general bugs, one decided (the place veto, off after two right and two wrong in four uses), the rest open at one of three.

4 Benchmark

Clips and labels. 274 clips were hand-labelled into four scenarios: *unseen_ambient* (an ambient sound whose source is not on screen), *mixed* (some sources visible, some not), *seen_ambient* (every ambient sound’s source is on screen) and *no_ambient* (speech or music only). The central labelling rule is the redundancy test used by professional captioners (DCMP): not “is the physical source in frame?” but “does the picture already tell the viewer that this sound is happening?” – wind roar over a rider filmed on a moving motorcycle is *seen*, and so are waves when the sea fills the frame. The main runs use 100 clips, 25 per scenario, so that abstention is scored on as many clips as augmentation.

Yield. The proposal named 300 clips. Four independent sourcing strategies (keyword scraping, AudioSet temporally-strong segments, curated film scenes, UnAV-100 overlap moments) and a three-stage automatic filter all landed in the same 15–20% band of usable positives, and yield fell as filtering grew. Yet 36% of clips that contain ambient sound at all are unseen or mixed: the phenomenon is not rare, the acquisition funnel discards it. The first 209 tags predate any model-based filtering and are the unbiased sample.

Label stability. Sixty tagged clips were replayed shuffled with the label hidden: self-agreement 78%, Cohen’s $\kappa = 0.60$. The instability is in the positive class (*unseen_ambient* reproduced 12/20 against *seen_ambient* 19/20), which is the class the gate is judged on.

Tuning on the test set, checked. The labelled clips were split 50/50 stratified by tag, every threshold swept on one half and frozen on the other, for five seeds: the held-out half scored higher than the sweep half in four of five, and the sweep chose the same corner of the grid each time. There is no optimism to remove, because the thresholds are not doing the work; the gate is limited by what a vision-language model can answer, not by its configuration.

4.1 External check: DCASE 2025 Task 3 on/off-screen labels

visibility check vs. DCASE 2025 gold (258 events)	
agreement	66.7%
off-screen recall (sound correctly not silenced)	97%
on-screen recall (visible source correctly silenced)	35%

The check is timid in one direction: it almost never silences a sound whose source is off screen, and silences only a third of those whose source is on screen. [DCASE geometric caveat: 360-degree capture; on-screen is defined by the field of view of the equirectangular frame.]

5 Evaluation protocol

5.1 Three references

model-derived (v2, v3) as specified in the proposal: written by the system’s own VLM from the detector’s own events. Found circular on 2026-09-14 (review): it rewards whichever system repeats the detector.

human-grounded (v2_grounded, v3_grounded) the same reference corrected by the annotator’s label; only computable on labelled clips; raises the abstaining system more than the baselines by arithmetic (notes §Main Result).

independent (v2_indep, v3_indep) built by models that never see the system’s output: Qwen2-Audio listens to the Demucs no-vocals stem and to the raw mix in 10 s chunks (the union, speech tags dropped; the stem alone was empty on a third of the clips because Demucs pulls tonal sounds into the vocals stem); each claim is verified by CLAP, kept if this clip’s audio ranks in the top 20 of the 100 benchmark clips for the claim’s text with positive cosine (a mean $+2\sigma$ bar over text decoys, the first form, left 58 of 100 references empty against 25 clips tagged *no_ambient*; both are reported); Idefics3-8B looks at the frames for what is on screen; Phi-3.5-mini writes the reference (Llama-3.1-8B is gated and was not accessible to the account); a MiniLM sentinel ($\tau = 0.50$, uncalibrated) rejects sentences unsupported by the verified list. This is the corrected primary.

5.2 Baselines

Blind audio-to-image (depict everything heard, no gate) and audio captioning (text instead of a picture); video reasoning off for both.

5.3 Statistics

Per-clip paired win/tie/loss, bootstrap 95% CI on the mean difference (5000 resamples), and an oracle-gate bound: the human tag as the gate, the best system’s output where a picture is due (`scripts/paired_stats.py`).

6 Results

6.1 Main comparison

run	reference	proposed
v2	model-derived	2.27
v2_grounded	human-grounded	2.90
v2_indep_list	independent (list-and-match visibility)	2.41
v3	model-derived	[pending job 28942456]
v3_grounded	human-grounded	[pending]
v3_indep	independent	[after v3 describe: reference phase, then judge]

6.2 Paired statistics

run	proposed vs. blind: win / tie / loss	mean diff	95% CI	oracle gate
v2	11 / 52 / 37	-0.70	[-0.98, -0.43]	2.89
v2_grounded	22 / 54 / 24	-0.24	[-0.47, -0.02]	3.19
v2_indep_list	12 / 59 / 29	-0.55	[-0.82, -0.30]	2.88
v3_*	[pending]			

Under every reference the blind baseline scores at least as high as the gated system on v2; the interval excludes zero each time. The circular reference was not what held the gate back. The oracle row is the ceiling any gate could reach on this benchmark with the best system’s output where a picture is due: 2.88–3.19, i.e. a perfect gate would beat the blind baseline by at most +0.05 under the human-grounded reference and not at all under the others.

The independent reference is not neutral either. Its silence decision (“nothing beyond the picture”) agrees with the human tag on 55 of 100 clips – chance. The list-and-match visibility step (Idefics3 asked for a list of sound-capable things, one word back per clip; MiniLM cosine to each verified sound) calls almost every sound unseen, so on seen-ambient clips the reference says the viewer would miss the sound and the blind baseline is rewarded for showing it (seen_ambient: blind 3.08, proposed 2.48). A per-sound question to the same VLM (“is the thing producing *X* visible? answer **visible:** <name> or not **visible**”) was tried next and scored the same 55%, because the rule “silent only if every verified claim is visible” fails whenever one of five to eight audio-LM claims (*video game sound*, *wind noise (microphone)*, *clock ticking*) is not; that variant was not judged (a 70% agreement bar gates the judge pass). The reference’s remaining weakness is therefore the same question the system itself finds hardest – whether a sound’s source is on screen – and the honest reading is that no automatic reference built here can score the gate decision; the human-grounded reference remains the only one that can, with the asymmetry already stated.

6.3 Per-category

[Per human tag table from `paired_stats.py` output, per reference.]

6.4 Onset timing

Against seven hand-labelled onsets on three clips the plain stamp was late by +0.9s on average and the shipping method by +0.3s; that is an anecdote, so the same question was put to DCASE 2025 Task 3, whose events carry a start frame at 100 ms. From 255 gold events (nine non-speech classes, onset inside the clip) one BEATs pass was stamped three ways; an event counts as

matched when a detection in the same AudioSet family starts within 1.5 s of it. Every error below is measured on the matched events only – 66 of 255 (26%) at the display bar – so the table describes where a picture is stamped when a sound is detected, not how often it is.

stamping	matched	mean signed	MAE	≤ 0.25 s	≤ 0.5 s
sliding-window stamp (before 09-14)	50	+0.29	0.43	44%	68%
+ hysteresis	66	+0.22	0.35	58%	74%
+ hysteresis + occlusion (shipping)	66	+0.16	0.35	45%	80%

Hysteresis is the gain: on the 50 events both methods match, MAE falls from 0.43 to 0.31 s, and it matches sixteen more events, because a sound that reaches the bar late is now traced back to where it began. Occlusion adds nothing on average: on the 65 events both match it is better on 25 and worse on 39 with an identical MAE of 0.34 s (paired bootstrap difference 0.00, 95% CI $[-0.11, +0.10]$). It trades a smaller late bias and more onsets within half a second for fewer within a quarter-second; on knocks, where hysteresis alone is within 0.08 s, it moves the onset +0.26 s late. The seven-clip number therefore overstated the method, and the honest statement is that the shipping onset is within half a second for four detected events in five. Detector recall on these five-second clips is the larger limit: laughter, footsteps and domestic sounds rarely reach 0.35 on BEATs. Whether to keep occlusion is a development-split decision left open (§3.6); it does not alter detection or gating, only where a picture is stamped.

6.5 Ablations recorded during development

[One short paragraph each, all numbers from the notes.]

7 Discussion

[RQ1–RQ5.]

8 Limitations

[From LIMITATIONS.md.]

9 Conclusion and future work

[Open-vocabulary audio (CLAP zero-shot / audio LLM) as a second opinion; source-separation front end; human study; second annotator; τ calibration of the sentinel on DCASE gold.]